

AB POWER ENGINEERING

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ELECTRIC MOTOR

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1. A device that converts electrical energy to mechanical energy to produce power.

- a. Battery
- b. Motor
- c. Generator
- d. None of the above

2. A device that uses mechanical power such as turbine, engine, etc. to produce electricity.

- a. Battery
- b. Motor
- c. Generator
- d. None of the above

3. A device that allows generator to feed current to a grid at the same time.

- a. Inverter
- b. Synchronizer
- c. Transformer
- d. None of the above

4. An inverse generator.

- a. Alternator
- b. Electric motor
- c. Internal combustion engine
- d. None of the above

5. A motor that can be powered by a 24- volt lead-acid battery.

- a. AC Motor
- b. DC Motor
- c. BC Motor
- d. All of the above

6. A term that indicates the maximum load that can be successfully carried by the motor if it is to be operated continuously and remain within a safe temperature range.

- a. Duty rating
- b. Phase
- c. Service factor
- d. None of the above

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7. A motor torque at zero speed or the maximum torque required to start the load.

- a. Twisting Torque
- b. Driving Torque
- c. Starting Torque
- d. None of the above

8. The type of motor with medium starting torque commonly used in industry.

- a. Split-phase motor
- b. Capacitance-type motor
- c. Repulsion-start motor
- d. None of the above

9. The type of motor with low starting torque.

- a. Split-phase motor
- b. Capacitance-type motor
- c. Repulsion-start motor
- d. None of the above

10. The type of motor with a very high starting torque.

- a. Split-phase motor
- b. Capacitance-type motor
- c. Repulsion-start motor
- d. None of the above

11. An alternator that does not need a voltage regulator in charging battery.

- a. DC-type alternator
- b. AC-type alternator
- c. IC-type alternator
- d. All of the above
- e. None of the above

12. A DC alternator basically rated in terms of voltage and

- a. watts
- b. ampere
- c. horsepower
- d. All of the above
- e. None of the above

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13. An alternator is rated 12 volts, 60 ampere. If two 100-AH battery is to be charged from 11 to 12 volts, how many hours the charging of the battery would be?

- a. 2.45 hours
- ☒ b. 3.33 hours
- c. 4.50 hours
- d. None of the above

14. Which of the following are the jobs of a motor?

- a. To drive the load at proper speed.
- b. To be able to start a load.
- c. To be able to withstand the abuse of the surroundings.
- ☒ d. All of the above
- e. None of the above

15. NEMA stands for ____.

- a. National Electric Motors Association
- ☒ b. National Electrical Manufacturers Association
- c. National Electrical Machinery Association
- d. None of the above

16. Based on NEMA standards, which of the following statements is true?

- a. The higher the motor power, the higher is the efficiency.
- b. The higher the efficiency of the motor, the higher is the power factor.
- c. The higher the motor power, the higher is the power factor.
- ☒ d. All of the above

17. A motor that is usually connected to the grid.

- ☒ a. AC Motor
- b. DC Motor
- c. Both AC & DC Motors
- d. None of the above

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18. The important information necessary for the proper application of the motor.

- a. Specification
- ☒ b. Nameplate
- c. Operators manual
- d. None of the above

19. The term that refers to how frequent the motor is used.

- ☒ a. Duty rating
- b. Phase
- c. Service factor
- d. None of the above

20. What size of motor would you recommend if 1-hp engine is to be replaced?

- ☒ a. 3/4-hp motor
- b. 1-hp motor
- c. 1.5-hp motor
- d. None of the above

21. The part of an electric motor that protects it from dust and water.

- a. Fan
- ☒ b. Enclosure
- c. Oil seal
- d. None of the above

22. The type of a motor enclosure that protects the motor from dangerous locations.

- a. Drip-proof enclosure
- b. Totally-enclosed enclosure
- ☒ c. Explosion-proof enclosure
- d. None of the above

23. The number of individual voltages applied to the motor.

- a. Ampacity
- b. Voltage Drop
- ☒ c. Phase
- d. All of the above
- e. None of the above

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24. A motor having three individual voltages applied to it.

- a. Single-phase motor
- b. Two-phase motor
- ☒ c. Three-phase motor
- d. None of the above

25. A motor having one voltage applied to it in the shape of a sine wave.

- ☒ a. Single-phase motor
- b. Two-phase motor
- c. Three-phase motor
- d. None of the above

26. Which of the following motors are classified based on the method of starting?

- a. Split-phase motor
- b. Capacitor-start induction motor
- c. Repulsion-start induction motor
- ☒ d. All of the above
- e. None of the above

27. A three-phase motor delivers 10-ampere current at 220 volt. What is the power output of the motor?

- a. 2.2 kW
- b. 2.9 kW
- c. 3.4 kW
- ☒ d. None of the above

28. Electric-driven motors, with total power input of 35 kW, are designed for a rice milling plant. What generator would you recommend for the rice mill.

- a. 100-kVA single-phase generator
- b. 50- kVA single-phase generator
- c. 35-kVA single-phase generator
- ☒ d. None of the above

29. The fan of a flatbed dryer requires 7.5 kW motor. What motor would you recommend to drive the fan?

- a. 7.5 kW single-phase motor
- ☒ b. 7.5 kW three-phase motor
- c. 7.5 kW two-phase motor
- d. None of the above

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30. A 1.5-kW split-phase motor running at 1740 rpm is to be used to drive a bucket elevator. The elevator should run at a speed of approximately 60 rpm. If the service factor for the bucket elevator is 1.3, what is the design power for the drive?

- a. 1.50 kW
- b. 1.95 kW
- c. 2.01 kW
- d. None of the above

31. What is the speed ratio for the drive in Item 30 if the motor pulley is 4 inches?

- a. 1: 20
- b. 1: 29
- c. 1: 38
- d. None of the above

32. Which of the following belt drives is suitable for the motor in Item 30?

- a. Type A
- b. Type B
- c. Type C
- d. None of the above

33. A 7.5-kW drive is required to drive a hammer mill. As an Agricultural Engineer, what type of motor would you recommend for this?

- a. Single-phase capacitor-type motor
- b. Two-phase capacitor-type motor
- c. Three-phase capacitor-type motor
- d. None of the above

34. If the motor is heavily loaded and the voltage input is low, the motor will be damaged because it will ____ and the wires of the motor will be overheated.

- a. decrease the input current
- b. increase the input current
- c. not change the current
- d. None of the above

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35. The percentage of the energy supplied to the motor that is converted into mechanical energy at the motor's shaft when the motor is continuously operating at full load with the rated voltage applied.

- a. Motor power factor
- b. Motor efficiency
- c. Motor operating load
- d. None of the above

36. A motor-driven blower is used to control the amount of air for the combustion of fuel. In order to maintain the temperature of drying air, what motor control would you recommend to switch ON and switch OFF the blower?

- a. Magnetic control switch
- b. AC relay switch
- c. Thermostat
- d. None of the above

37. The armature winding or the rotating part of an electric motor which is typically constructed of a laminated steel core containing current-carrying copper wires.

- a. Rotor
- b. Contactor
- c. Stator
- d. None of the above

38. What would you recommend if an AC motor is to be powered by a battery?

- a. Connect directly the motor to the battery.
- b. Provide an inverter between the battery and the motor.
- c. Connect a step-up transformer between the battery and the motor.
- d. None of the above

39. The speed range of a four-pole electric motor.

- a. 800 to 1200 rpm
- b. 1600 to 1800 rpm
- c. 2000 to 3000 rpm
- d. All of the above

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40. The speed range of an eight-pole electric motor.

- a. 800 to 1200 rpm
- b. 1600 to 1800 rpm
- c. 2000 to 3000 rpm
- d. All of the above

41. A motor is designed to drive a 5-kW hammer mill. The motor will run at 1740 rpm speed on a 220-volt line. What size of breaker switch would you recommend to protect the motor? Assume a $0.8 \cos \theta$ for the motor.

- a. 15 amp
- b. 20 amp
- c. 30 amp
- d. None of the above

42. The size of wire used for the motor for starting is usually ____ than the wire used for running.

- a. smaller
- b. equal
- c. larger
- d. None of the above

43. The connecting and disconnecting switch for the alternator and battery should be located at the ____ terminal wire connecting the battery and the alternator.

- a. negative
- b. positive
- c. neutral
- d. All of the above
- e. None of the above

44. A field pole or the stationary part of an electric motor consisting of copper windings which is placed in a laminated iron core.

- a. Rotor
- b. Contactor
- c. Stator
- d. None of the a

45. Electrical wire system commonly used in multipass rice mill.

- a. Single-phase
- b. Three-phase
- c. Four-phase
- d. None of the above

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46. Motor frequency used in the Philippines.

- a. 50 Hz
- b. 60 Hz
- c. 70 Hz
- d. D. All of the above

47. Wire connection provided for electrical motor to provide safety and fault protection.

- a. Ground wire
- b. Neutral wire
- c. Feeder wire
- d. None of the above

48. A conductor that is intentionally connected to the neutral point of the system's power source to provides a return path for unbalanced electrical loads, allowing for the use of both three-phase and single-phase equipment.

- a. Ground wire
- b. Neutral wire
- c. Feeder wire
- d. None of the above

49. Commonly use power factor in estimating the power that can be delivered by a transformer.

- a. 0.60
- b. 0.70
- c. 0.80
- d. None of the above

48. If a three-phase motor with 60 Hz is to be connected to a 50 Hz line, what will happened to the speed of the motor?

- a. It will decrease
- b. It will increase
- c. It will rotate at the same speed
- d. None of the above

49. Three-phase motor that is commonly used in multi-pass rice mill usually operate on ____ volt.

- a. 220
- b. 360
- c. 440
- d. None of the above

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50. Converter used in converting single-phase line to supply power to three-phase motor and is suitable for application with changing load.

- a. Rotary Phase Converter
- b. Digital Phase Converters
- c. Variable Frequency Drive
- d. All of the above

51. This is a three-phase converter that uses a single-phase motor, known as an idler motor, to create the third phase.

- a. Rotary Phase Converter
- b. Static Phase Converter
- c. Digital Phase Converter
- d. None of the above

52. This converter use capacitors to produce a synthetic three-phase output from a single-phase input and is best for applications where motor doesn't start and stop frequently or change speed.

- a. Rotary Phase Converter
- b. Static Phase Converter
- c. Digital Phase Converter
- d. None of the above

53. A modern digital phase converters that uses digital signal processors and power electronics to create a stable, balanced three-phase power supply and are ideal for running precision equipment.

- a. a. Rotary Phase Converter
- b. Static Phase Converter
- c. Digital Phase Converter
- d. None of the above

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GOD BLESS !!!

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